



Debug Detective

Quick facilitation notes & answer key

Goal

The unit finale: students debug a longer program (find the wrong-way block, fix it, explain why the fix works), then create their own buggy program for a partner. Reading and altering code and describing how the change fixes the outcome is C3.2; writing their own program (with a target) is C3.1.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 with a marker on 4 so students can run the mystery program and test their own.

Run it (about 20-25 min)

1. I DO: run the mystery program — 'start 0, forward 2 to 2, back 1 to 1, forward 1 to 2. Target is 4 — the back 1 sent Bit the wrong way.'
2. WE DO: Exercise 1 — Bit lands on 2. Exercise 2 — circle 'back 1', swap it to 'forward 1'; the program reaches 4 (0 → 2 → 3 → 4).
3. YOU DO: Exercise 3 — students explain the fix (a back block was turning Bit around) and write their own one-bug program with a target.
4. Celebrate: 'You can find and fix every kind of bug — you are a debugger!'

Answer key (modelled by the run-gate)

Ex 1 — mystery program: 0 → forward 2 → 2 → back 1 → 1 → forward 1 → 2. Bit lands on 2, not the target 4.

Ex 2 — the wrong-way block is 'back 1'; swap it to 'forward 1': 0 → 2 → 3 → 4. Bit reaches the target, 4.

Ex 3 — explanation: the back 1 block moved Bit away from the target; changing it to forward 1 keeps Bit going toward 4. Make-your-own is open; a sample is start 0, forward 1, forward 1 with target 3 (a missing-block bug — add one more forward 1).

Watch for & differentiate

Common slip: changing more than one block — only one block is wrong; the smallest fix is to swap the back 1.

Simplify: tell students the bug is a wrong-way block and have them find which one.

Extend: have partners trade their make-your-own programs and time how fast they each debug.

Success indicator: the child lands the mystery program on 2, swaps back 1 to forward 1 to reach 4, explains the fix, and writes a solvable one-bug program.